

SATNET EDU / Lecture 05

# Designing and Evaluating Satellite Networks

How to turn a network idea into a controlled, reproducible experiment.

## LECTURE 05

From a design question to an explained result

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State the question. Control the comparison. Inspect the outcome.

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# Start with a Question That the Model Can Answer

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Connect a design choice to a measurable outcome under stated conditions.

► **Turn a goal into a comparison.**

- Ask how a specific topology change affects connectivity between two chosen ground sites.

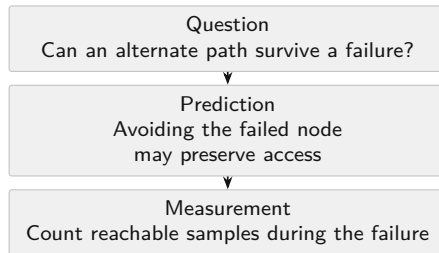
► **Predict a mechanism first.**

- Allowing more candidate ISLs may create alternate paths. Range and Earth blockage can still remove those links.

► **Match the claim to the model.**

- SatNet Edu can compare geometric reachability and propagation. It cannot predict throughput without a traffic model.

## Make the question testable



Fix endpoints, time window, geometry, and routing metric

The teaching model omits antenna scheduling, capacity, queues, packet loss, and routing convergence.

# Change One Factor and Keep the Rest Comparable

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A controlled comparison helps identify why a measured result changed.

► **Use a shared baseline.**

- Use the same epoch, constellation, sites, elevation mask, duration, and sample times in both runs.

► **Name the factor you change.**

- Here the failure schedule is the treatment. Add one satellite outage while leaving geometry and routing settings fixed.

► **Separate follow-up questions.**

- A later experiment can change topology or altitude. Changing both at once makes a single-cause explanation harder.

## The paired failure experiment

| Setting         | Baseline             | Failure run          |
|-----------------|----------------------|----------------------|
| Satellites      | $6 \times 12$        | $6 \times 12$        |
| Altitude / tilt | 1000 km / $53^\circ$ | 1000 km / $53^\circ$ |
| Access mask     | $10^\circ$           | $10^\circ$           |
| ISL limit       | 4000 km              | 4000 km              |
| Window / step   | 20 min / 10 s        | 20 min / 10 s        |
| Outage          | None                 | 600–900 s            |

Vancouver  $\longleftrightarrow$  Tokyo   Same epoch

Companion experiment uses orbital-neighbor links and epoch 2026-01-01 UTC. The node selection rule is documented.

# Coverage, Reachability, and Delay Need Different Counts

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Always state the population used in a metric, especially when some samples have no route.

► **Coverage is local to one site.**

- Count samples with at least one usable ground link. This does not require a route to the destination.

► **Reachability is a pairwise property.**

- Count samples with a complete source-to-destination path, using all scheduled observation times as the denominator.

► **Delay is conditional on a route.**

- Average valid delays only and report the reachability fraction alongside them. Missing paths are not zero-delay paths.

**The denominator changes the meaning**

| Time | U access | U-V route | Delay |
|------|----------|-----------|-------|
| 0 s  | Yes      | Yes       | 12 ms |
| 10 s | Yes      | No        | —     |
| 20 s | Yes      | Yes       | 18 ms |
| 30 s | No       | No        | —     |

Coverage at U:  $3/4 = 75\%$

Pair reachability:  $2/4 = 50\%$

Conditional delay:  $(12 + 18)/2 = 15$  ms

Constructed four-sample example. Delay here means one-way propagation. These are sample fractions, not exact time fractions.

# Unequal Sample Spacing Requires Time Weights

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Extra samples near an event can change a sample average without changing the time history.

► **Samples can represent unequal time.**

- A recorder can add samples at failure times. Counting every observation equally can overweight short periods.

► **Choose and label an estimator.**

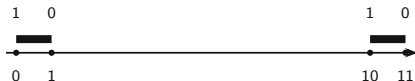
- A left-hold estimate assumes the observed state lasts until the next sample. Weight each state by that interval length.

► **Check sensitivity to finer sampling.**

- This assumption can miss an unobserved change. Report the interval, window, and approximation with the metric.

## Two reachable samples, unequal durations

$a$ : 1 for reachable, 0 otherwise



Sample fraction:  $2/4 = 50\%$

Held-state time fraction:  $(1 + 1)/11 = 18.2\%$

Time in seconds. Final sample has zero following duration.

Constructed states at 0, 1, 10, and 11 s. The time fraction assumes the pictured state is held between samples.

# A Failure Test Should Target a Defined Component

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A failure experiment is meaningful only when the disabled component and its timing are specified.

► **Choose a repeatable selection rule.**

- Select an interior satellite on the baseline route at 600 seconds. Save its ID so the stress case can be reproduced.

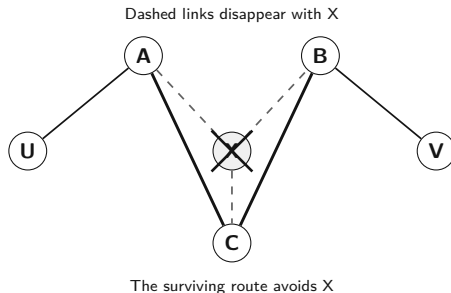
► **Remove every incident link.**

- A disabled satellite cannot forward traffic. Removing just one displayed route edge would test a different fault.

► **Look for a surviving alternate path.**

- The replacement can be longer, or no path may remain. Neither outcome should be assumed before querying the graph.

## One failed node affects multiple links



Constructed graph. A targeted route-node failure tests a stress case, not average fleet reliability.

# Compare the Failed Run with Its Matched Baseline

Orbital motion continues during a failure, so before-and-after values alone can mix two effects.

## ► Compare the same timestamps.

- At each timestamp, compare both runs under identical motion. Their difference isolates the scheduled failure.

## ► Separate outage from extra delay.

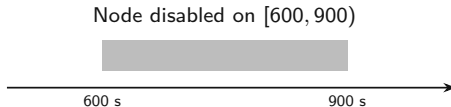
- Calculate delay differences only where both runs have a route. Report additional unreachable samples separately.

## ► Inspect recovery independently.

- Restoring a node allows links to return. The model recomputes routes immediately, without protocol recovery delay.

## A matched comparison during one event

| Time  | Baseline | Failure | Difference |
|-------|----------|---------|------------|
| 590 s | 12 ms    | 12 ms   | 0 ms       |
| 600 s | 13 ms    | 18 ms   | +5 ms      |
| 700 s | 14 ms    | No path | —          |
| 900 s | 15 ms    | 15 ms   | 0 ms       |



Constructed comparison values. The companion script supplies actual traces and metrics for the chosen scenario.

# Keep Designs That Offer a Useful Tradeoff

No single metric captures satellite count, connectivity, and path quality at the same time.

► **Choose a small set of objectives.**

- Here we prefer fewer satellites, a higher reachable-sample fraction, and lower conditional propagation delay.

► **Reject a dominated design.**

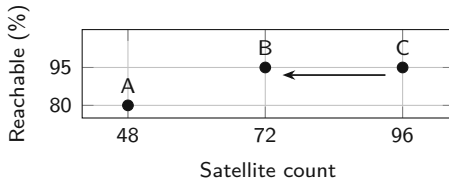
- B beats C on satellite count and delay while matching reachability. C offers no benefit under these chosen metrics.

► **Keep useful alternatives.**

- A is cheaper but less connected than B. Choosing between them requires an explicit service goal or resource budget.

## An illustrative three-design comparison

| Design | Satellites | Reachable | Delay |
|--------|------------|-----------|-------|
| A      | 48         | 80%       | 24 ms |
| B      | 72         | 95%       | 20 ms |
| C      | 96         | 95%       | 25 ms |



Constructed results, not simulator output. Satellite count is a resource proxy, not a complete financial cost.

# Save the Scenario and the Recorded Outcomes

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A reproducible result needs more than a screenshot of a favorable instant.

► **Preserve inputs and selection rules.**

- Record the epoch, orbit parameters, topology, endpoints, failure ID, and sampling schedule with the script.

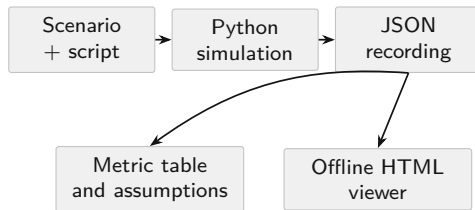
► **Keep the complete recording.**

- Save states and routes, including no-path samples. A viewer can replay the recording without running Python again.

► **State the model and time window.**

- Include dependency versions, metrics, and limitations. A short synthetic run does not establish worldwide operational service.

## A result another student can inspect



The viewer displays records. It does not recompute routes.

The exported trace includes scenario data and dependency versions. The basic model is deterministic.

# Build a Defensible Satellite-Network Experiment

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Combine architecture, coverage, routing, and time-varying behavior in a small final investigation.

► **Begin with the paired failure script.**

- Predict the outcome, run both cases, and locate any extra disconnections or detours in the viewer.

► **Make one justified design revision.**

- Change the topology policy or constellation layout. Repeat the same failure rule and compare the same metrics and sites.

► **Explain a result and a limitation.**

- Use a route diagram and a time plot to explain the mechanism. State what the experiment cannot conclude.

Use the same failure ID to isolate a design change. Retargeting a new route node asks a different stress-test question.

## A compact final submission

| Include               | Answer                          |
|-----------------------|---------------------------------|
| Question + prediction | What should change, and why?    |
| Baseline + revision   | Which design factor changed?    |
| Saved JSON + HTML     | Can another student inspect it? |
| Metrics + time plot   | How often, how long, how far?   |
| Conclusion + limits   | What does this run support?     |

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python teaching/lecture-05/experiment.py
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Outputs: paired traces, viewers, and CSV metrics